

LAB 5: UNDERSTANDING ETHERNET AND ARP USING WIRESHARK

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SECTION: C

PART A

1. Provide a screenshot of the packet that contains the HTTP GET request sent to the server, including the Ethernet frame information.

http						
No.	Time	Source	Destination	Protocol	Length	Info
276	8.392748...	34.107.221.82	10.1.22.244	HTTP	364	HTTP/1.1 200 OK (text/html)
804	8.941055...	10.1.22.244	34.107.221.82	HTTP	376	GET /success.txt?ipv4 HTTP/1.1
816	8.967230...	34.107.221.82	10.1.22.244	HTTP	282	HTTP/1.1 200 OK (text/plain)
1813	20.55977...	10.1.22.244	128.119.245.12	HTTP	443	GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1
1825	20.88496...	128.119.245.12	10.1.22.244	HTTP	2034	HTTP/1.1 200 OK (text/html)

Frame 1813: Packet, 443 bytes on wire (3544 bits), 443 bytes captured (3544 bits) on interface wlan0, id 0

Ethernet II, Src: Intel_24:79:ad (d4:f3:2d:24:79:ad), Dst: HewlettPacka_4a:d7:80 (3c:a8:2a:4a:d7:80)

Destination: HewlettPacka_4a:d7:80 (3c:a8:2a:4a:d7:80)

.....0. = LG bit: Globally unique address (factory default)

.....0. = IG bit: Individual address (unicast)

Source: Intel_24:79:ad (d4:f3:2d:24:79:ad)

.....0. = LG bit: Globally unique address (factory default)

.....0. = IG bit: Individual address (unicast)

Type: IPv4 (0x0800)

[Stream index: 11]

Internet Protocol Version 4, Src: 10.1.22.244, Dst: 128.119.245.12

Transmission Control Protocol, Src Port: 36568, Dst Port: 80, Seq: 1, Ack: 1, Len: 377

Hypertext Transfer Protocol

2. In this packet, what is the Ethernet source address? Is this the physical (MAC) address of your

device?

- > Ethernet source address: d4:f3:2d:24:79:ad
- > Yes. This is the physical (MAC) address of my device (NIC).

3. In the same packet, what is the Ethernet destination address? Is this the physical address of the server? If not, provide a justification.

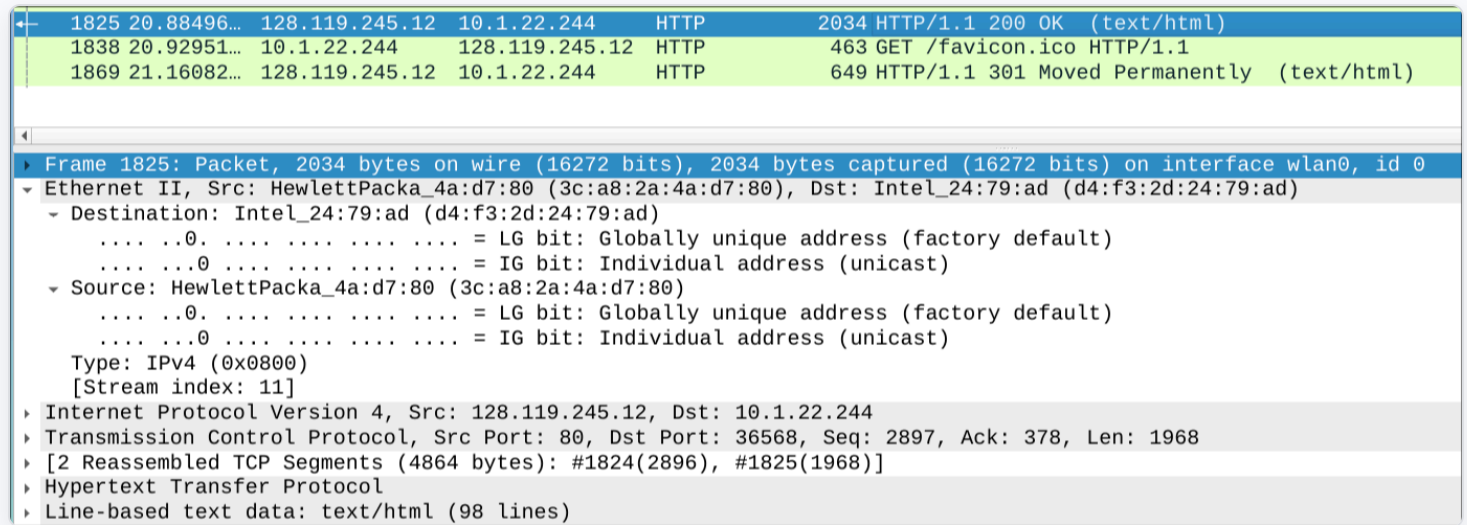
- > Ethernet destination address: 3c:a8:2a:4a:d7:80
- > No. This is NOT the server's MAC address.
- > Ethernet frames operate only within the local network (LAN). Since the server (128.119.245.12) is on a different network, device sends the frame to the next hop , typically the default

gateway (router). Therefore, this MAC address belongs to the router, not the remote server.

4. What is the destination IP address in this packet? Does it correspond to the IP address of the server?

- > Destination IP address: 128.119.245.12
- > Yes. This corresponds to the IP address of the server.
- > IP operates end-to-end across networks, unlike Ethernet. So even though the Ethernet frame is sent to the router, the IP packet inside still carries the final destination (the server).

5. Provide a screenshot of the packet containing the HTTP text response from the server, including the Ethernet frame information.



6.

- > In the earlier HTTP GET packet:
 - Ethernet Source Address = device's MAC address
 - Ethernet Destination Address = router's (default gateway) MAC address
- > In this packet (HTTP response):
 - Ethernet Source Address = router's MAC address
 - Ethernet Destination Address = device's MAC address

- > The direction of communication is reversed.
- > For the request:
 - DEVICE → ROUTER → INTERNET → SERVER
- > For the response:
 - SERVER → ROUTER → DEVICE
- > At the Ethernet layer, communication is limited to the local network (single hop). So the router rewrites the Ethernet frame before forwarding it to the device:
 - Source MAC = router's interface MAC
 - Destination MAC = device's MAC
- > Ethernet addresses change at each hop and represent only the local link, not the end-to-end path.

PART B:

1. Provide a screenshot of the ARP probe packet, clearly showing the sender IP address and target IP address. Explain why this packet is transmitted.

No.	Time	Source	Destination	Protocol	Length	Info
8	0.03841...	Intel_24:79...	Broadcast	ARP	42	Who has 10.1.22.244? (ARP Probe)
9	0.04338...	Intel_00:01...	Intel_24:79...	ARP	60	10.1.18.78 is at e4:c7:67:00:01:6d
Frame 8: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface wlan0, id 0 Ethernet II, Src: Intel_24:79:ad (d4:f3:2d:24:79:ad), Dst: Broadcast (ff:ff:ff:ff:ff:ff) - Destination: Broadcast (ff:ff:ff:ff:ff:ff) - Source: Intel_24:79:ad (d4:f3:2d:24:79:ad) - Type: ARP (0x0806) [Stream index: 4] Address Resolution Protocol (ARP Probe) - Hardware type: Ethernet (1) - Protocol type: IPv4 (0x0800) - Hardware size: 6 - Protocol size: 4 - Opcode: request (1) [Is probe: True] - Sender MAC address: Intel_24:79:ad (d4:f3:2d:24:79:ad) - Sender IP address: 0.0.0.0 - Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00) - Target IP address: 10.1.22.244						

› Sender IP address: 0.0.0.0

› Target IP address: 10.1.22.244

› This packet is transmitted to check whether the IP address is already in use on the network before assigning it to the device.

› This process helps avoid IP address conflicts. Since the device does not yet have an assigned IP address, it uses 0.0.0.0 as the sender IP.

2. Provide a screenshot of the ARP packet sent from your device, including the sender IP address and target IP address.

28	0.28301...	Intel_24:79...	Broadcast	ARP	42	Who has 10.1.16.1? Tell 10.1.22.244
29	0.28640...	HewlettPack...	Intel 24:79...	ARP	60	10.1.16.1 is at 3c:a8:2a:4a:d7:80
Frame 28: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface wlan0, id 0						
Ethernet II, Src: Intel_24:79:ad (d4:f3:2d:24:79:ad), Dst: Broadcast (ff:ff:ff:ff:ff:ff)						
› Destination: Broadcast (ff:ff:ff:ff:ff:ff)						
› Source: Intel_24:79:ad (d4:f3:2d:24:79:ad)						
Type: ARP (0x0806)						
[Stream index: 4]						
Address Resolution Protocol (request)						
Hardware type: Ethernet (1)						
Protocol type: IPv4 (0x0800)						
Hardware size: 6						
Protocol size: 4						
Opcode: request (1)						
Sender MAC address: Intel_24:79:ad (d4:f3:2d:24:79:ad)						
Sender IP address: 10.1.22.244						
Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Target IP address: 10.1.16.1						

› Sender IP address: 10.1.22.244

› Target IP address: 10.1.16.1

A. What differences do you observe when compared with the ARP probe packet?

› In the ARP probe:

- Sender IP address = 0.0.0.0
- Used for checking IP availability

› In the ARP request:

- Sender IP address = actual assigned IP (10.1.22.244)
- Used to discover the MAC address of another device

› The ARP probe is used before IP assignment, whereas the ARP request is used after the device has a valid IP.

B. What is the target MAC address in this packet?
Justify your answer.

› Target MAC address: 00:00:00:00:00:00

› This is because the device does not know the MAC address of the target IP.

› Therefore, the request is broadcast to all devices in the network to identify the correct MAC address.

C. Which opcode value in the packet indicates that it is an ARP request?

› Opcode value: 1

› This indicates that the packet is an ARP request.

3. Provide a screenshot of the ARP response packet received after the ARP request, showing the sender and target IP addresses.

No.	Time	Source	Destination	Protocol	Length	Info
29	0.28640...	HewlettPack...	Intel_24:79...	ARP	60	10.1.16.1 is at 3c:a8:2a:4a:d7:80
42	0.33126...	Intel_3e:bd...	Intel_24:79...	ARP	60	10.1.22.80 is at 80:e4:ba:3e:bd:81

Frame 29: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface wlan0, id 0

Ethernet II, Src: HewlettPacka_4a:d7:80 (3c:a8:2a:4a:d7:80), Dst: Intel_24:79:ad (d4:f3:2d:24:79:ad)

Destination: Intel_24:79:ad (d4:f3:2d:24:79:ad)

Source: HewlettPacka_4a:d7:80 (3c:a8:2a:4a:d7:80)

Type: ARP (0x0806)

[Stream index: 6]

Padding: 00

Address Resolution Protocol (reply)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: reply (2)

Sender MAC address: HewlettPacka_4a:d7:80 (3c:a8:2a:4a:d7:80)

Sender IP address: 10.1.16.1

Target MAC address: Intel_24:79:ad (d4:f3:2d:24:79:ad)

Target IP address: 10.1.22.244

- › Sender IP address: 10.1.16.1
- › Target IP address: 10.1.22.244

A. What changes do you observe compared to the ARP request packet?

- › In the ARP request:
 - Packet is broadcast
 - Sender asks for MAC address
- › In the ARP reply:
 - Packet is unicast (sent directly to the device)
 - Sender provides its MAC address
- › The roles are reversed: the requester becomes the receiver, and the target responds.

B. What is the target MAC address in this packet?
Justify your answer.

- › Target MAC address: d4:f3:2d:24:79:ad
- › This is the MAC address of the device.
- › Since the sender already knows the requester, the reply is sent directly (unicast) to the device.

C. Which opcode value indicates that the packet is an ARP reply?

- › Opcode value: 2
- › This indicates that the packet is an ARP reply.